Randomization Testing

Jericho Lawson

MATH 240

Week 9, Lecture #2 — Ch. 11

Steps of Hypothesis Testing

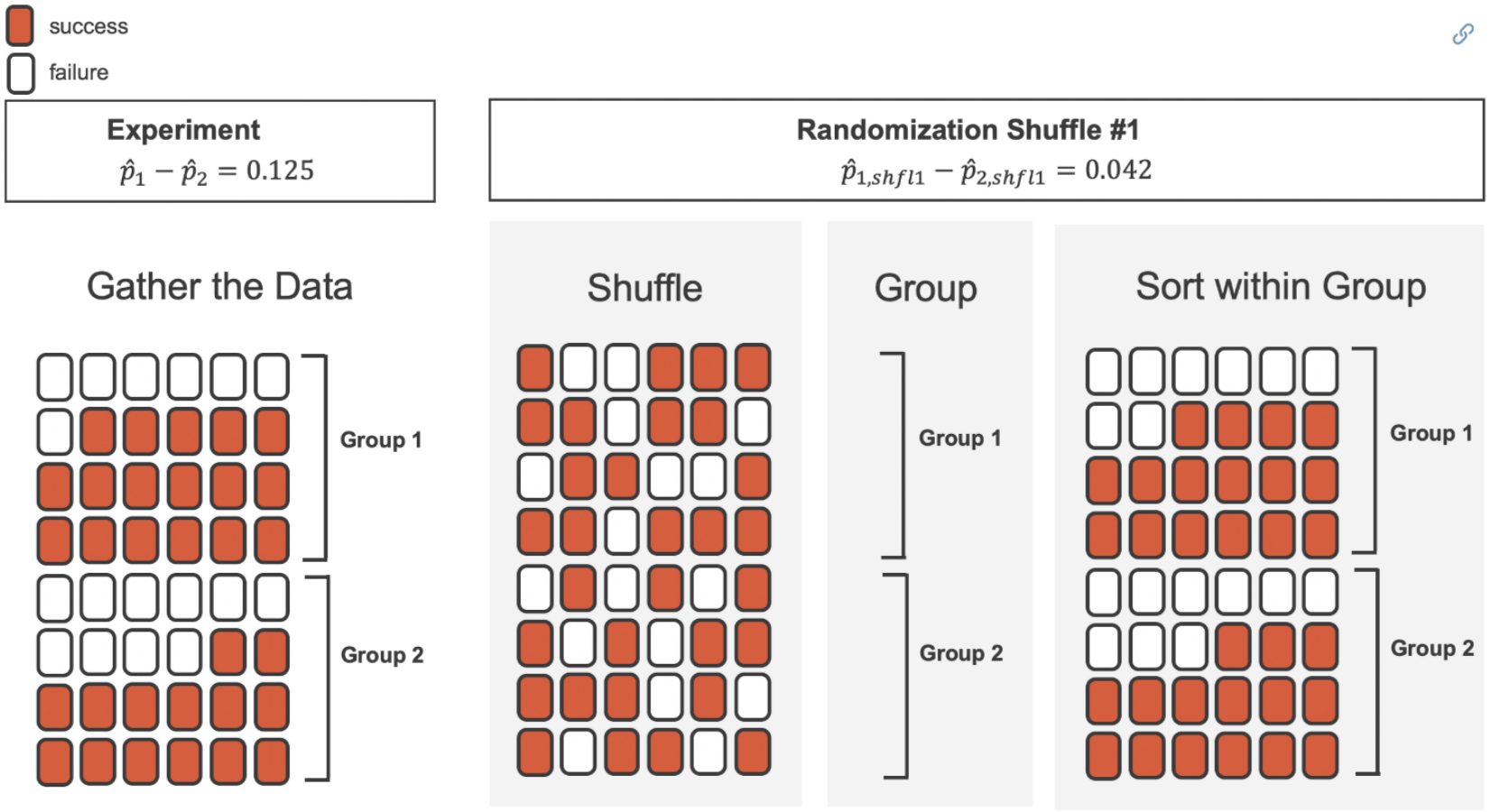
1. Define the problem
   - Parameter of interest
   - Null and alternative hypotheses
   - Level of significance
2. Collect data or run experiment
   - Determine sample statistic
3. Choose a model for testing
   - Appropriate test/model
   - Verifying conditions
4. Perform test/model
   - e.g. test statistic to get p-values and/or confidence intervals
   - e.g. randomization testing
5. Draw conclusions
   - Make decision on null/alternative hypotheses
   - Interpret decision in context of the problem

Randomization Testing

In many circumstances, we might not have a definitive idea for the null distribution, primarily in experiments where we have control and treatment groups.

- **Randomization test**: Shuffles the explanatory variable to mimic the natural variability found in a randomized experiment
  - If the null is true, we should expect no difference between the control and treatment groups
  - This implies that randomizing their assignments will not change the outcome
    - ★ If we truly have a noticable difference: this should not be as common of a difference compared to the randomization

Randomization Testing



Randomization Testing: What We're Doing

- Re-randomizing subjects to the control and treatment groups some N times
  - The higher N is, the better
- Gather sample statistics from both groups to get test statistic
  - i.e. difference in proportions
  - Results in sampling distribution of results
- Compare our observed difference against the null sampling distribution to calculate p-value

Is Yawning Contagious?

An experiment conducted by the MythBusters, a science entertainment TV program on the Discovery Channel, tested if a person can be subconsciously influenced into yawning if another person near them yawn. 50 people were randomly assigned to two groups:

- 34 to a group where a person near them yawned (treatment)
- 16 to a group where there wasn't a person yawning near them (control)

Is Yawning Contagious?

An experiment conducted by the MythBusters, a science entertainment TV program on the Discovery Channel, tested if a person can be subconsciously influenced into yawning if another person near them yawn. 50 people were randomly assigned to two groups:

- 34 to a group where a person near them yawned (treatment)
- 16 to a group where there wasn't a person yawning near them (control)

We will test some hypotheses based on the data!

HT: Is Yawning Contagious?

Step 1: Define the problem!

- Parameter of interest:
  - p_T: true probability of yawning in treatment group
  - p_C: true probability of yawning in control group

HT: Is Yawning Contagious?

Step 1: Define the problem!

- Parameter of interest:
  - p_T: true probability of yawning in treatment group
  - p_C: true probability of yawning in control group
- Null and alternative hypotheses:
  - Yawning is not contagious $\implies$ $H_0 : p_T = p_C$
  - Yawning is contagious $\implies$ $H_a : p_T > p_C$

HT: Is Yawning Contagious?

Step 1: Define the problem!

- Parameter of interest:
  - p_T: true probability of yawning in treatment group
  - p_C: true probability of yawning in control group
- Null and alternative hypotheses:
  - Yawning is not contagious $\implies$ $H_0 : p_T = p_C$
  - Yawning is contagious $\implies$ $H_a : p_T > p_C$
- Level of significance
  - We will choose a level of significance $\alpha = 0.01$.

HT: Is Yawning Contagious?

Step 2: Collect data or run experiment:

- Determine sample statistic
  - We will use $\hat{p}_T$ and $\hat{p}_C$, the sample proportions of those that yawn in the treatment and control groups respectively
  - The difference $\hat{p}_T - \hat{p}_C$ will constitute our sample statistic

$$\hat{p}_T = \frac{10}{34} = 0.2941$$

$$\hat{p}_C = \frac{4}{16} = 0.25$$

  - Sample statistic: $\hat{p}_T - \hat{p}_C = 0.0441$

HT: Is Yawning Contagious?

Steps 3/4: Choose a model for testing and perform the test:

- Use of a randomization test
- Let's look at the R code

HT: Is Yawning Contagious?

Step 4: Performing the Test

1. Randomly shuffle all participants into control and treatment groups (with correct number of participants)
2. Determine sample statistic in each of the groups; calculate difference (as test statistic)
3. Perform over and over for N samples
4. Calculate p-value, which is:

$$p\text{-value} = \frac{\#\text{ of samples whose sample statistic is more extreme}}{\#\text{ of total samples}}$$

HT: Is Yawning Contagious?

$$p\text{-value} = \frac{\#\text{ of samples whose sample statistic is more extreme}}{\#\text{ of total samples}}$$

More extreme means...

- Greater than or equal to test statistic if it's a right-sided test (e.g. H_a has $>$)
- Less than or equal to test statistic if it's a left-sided test (e.g. H_a has $<$)
- Absolute difference from common belief is greater if it's a two-sided test (e.g. H_a has $\neq$)

HT: Is Yawning Contagious?

Step 5: Draw Conclusions

- Say we get a p-value of 0.251
  - 25.1% of null samples have differences that are more extreme than what we observe
- With the p-value greater than α, we fail to reject H_0
- Not enough evidence to say that yawning is contagious

In Summary

(1) Define problem:

- $p_T - p_C$ is of interest, $\alpha = 0.01$

$$H_0 : p_T = p_C$$
$$H_a : p_T > p_C$$

(2) Collect data:

$$\hat{p}_T - \hat{p}_C = 0.0441$$

(3/4) Choose and perform model/test:

- Randomization testing, p-value is 0.251

(5) Draw conclusions:

- Fail to reject H_0 since $0.251 > 0.05$
- Not enough evidence to say yawning is contagious

Activity: Make Your Own Randomization Test

Two houses in a neighborhood are giving out candy for Halloween, and rumor has it that the second house is giving out more good candy than the first house.

- 20 kids are sent to each house and we get the following results:
  - $\hat{p}_1 = 0.4$
  - $\hat{p}_2 = 0.7$

Perform a hypothesis test using randomization testing with a discernibility level of 0.05.